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A paleomagnetic study of Piedmont metamorphic rocks in northern Delaware

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ABSTRACT

It has been suggested that during the lower Paleozoic the Appalachian Piedmont Province formed not as part of North America, but on the south or east side of the Iapetus Ocean. The results of a previous paleomagnetic study of the Arden pluton from the Piedmont of northern Delaware are in agreement with this hypothesis. For the present study, several metamorphic and igneous rocks were sampled across a wide area of the northern Delaware Piedmont. The Late Ordovician poles determined in this study, like the Arden pluton pole, lie close to their sites and do not agree with Ordovician paleopoles for cratonic North America. Tilting of the sampled rocks after the acquisition of magnetization is considered unlikely because the same steep direction is found in rocks collected over a wide area. The interpretation of these results is that the Piedmont of northern Delaware formed as part of the African plate and later became sutured to North America when the Iapetus Ocean closed and Pangea was assembled.

INTRODUCTION

D. W. Rankin (1975) has proposed that part of the Appalachian Piedmont Province, the migmatitic Inner Piedmont, was once part of the African plate and later became sutured to the North American plate during their collision as Pangea was assembled. In Rankin's (1975) model, the Pine Mountain belt, the Sauratown Mountains anticlinorium, and the Baltimore Gneiss domes are thought to be parts of the North American plate exposed in windows in the allochthonous Inner Piedmont. A paleomagnetic study by Rao and Van der Voo (1980) of the Arden pluton of the Wilmington Complex, which is located in the Piedmont Province of northern Delaware, supports Rankin's hypothesis. The magnetic direction of Late Ordovician age found for this pluton (Fig. 1) is quite divergent from North American Ordovician directions and is best explained if the Delaware Piedmont did not form as part of the North American plate, but rather as part of a different early Paleozoic plate on the south or east side of the proto-Atlantic (Iapetus) Ocean, most likely the African plate.

The purpose of this study is to further test Rankin's hypothesis by better defining the pole position for the rocks of the Piedmont of northern Delaware and to examine the possibility that the Arden pluton has been tilted by sampling a wide area adjacent to it. The reason for sampling a wide area is that in an igneous-metamorphic terrane such as the Appalachian Piedmont, no structural corrections can be made. As a result, unresolvable rotations of the measured magnetic direction due to tilting of the body after acquisition of the magnetization are always a possibility. However, if the same magnetic direction is found over an area too large to have likely tilted as a unit, then it can be reasonably assumed that the direction of magnetization has not been rotated to a large degree. From 40 sites in northern Delaware, southeastern Pennsylvania, and northeastern Maryland (Fig. 2), 354 specimens were collected and progressively demagnetized, using the alternating-field or thermal method. An orthogonal demagnetization diagram (Zijderveld, 1967) was constructed for each specimen in order to analyze the directions of magnetization.

GEOLOGIC SETTING

The geology of the northern Delaware area has been described by Ward and Groot (1957), Ward (1959), and Wagner and Crawford (1975) and mapped by Woodruff and Thompson (1972, 1975).

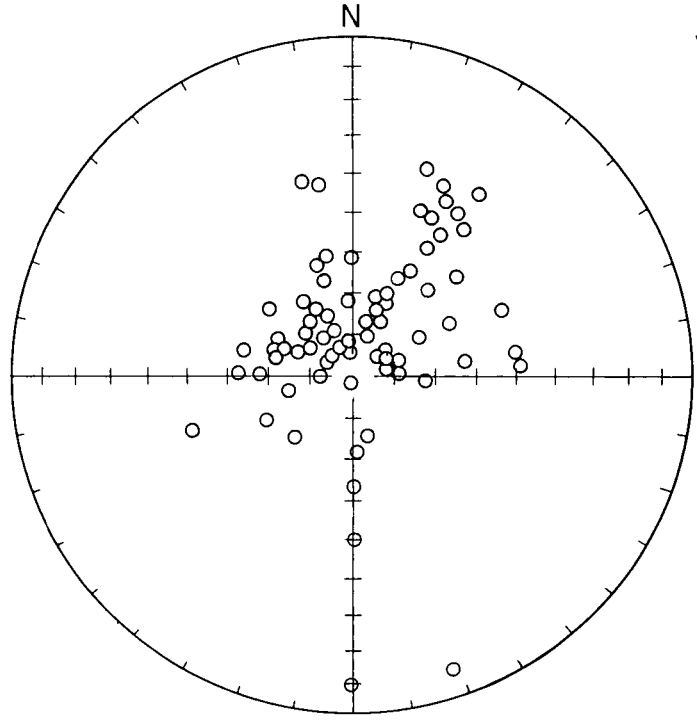


Figure 1. Equal-area stereographic projection of directions obtained from alternating-field demagnetization of 76 specimens from the Arden pluton (Rao and Van der Voo, 1980). All directions are upward.

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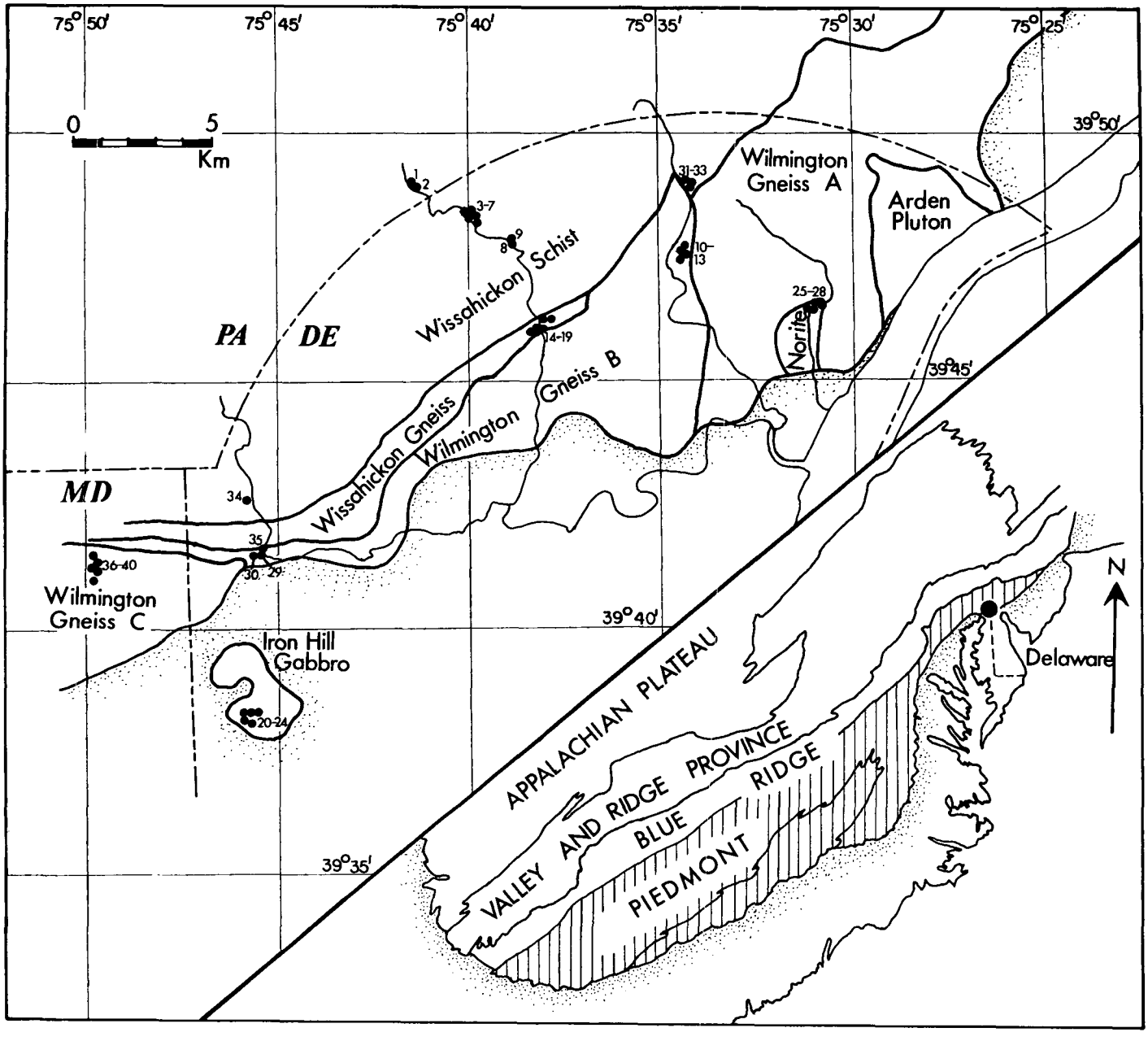


Figure 2. Generalized geologic map of northern Delaware and adjoining parts of Maryland and Pennsylvania, with site locations. Location map shows regional tectonic setting of study area.

In the northernmost part of the state, crystalline rocks of early Paleozoic or older age are exposed. These rocks constitute part of the Piedmont geologic, physiographic, and tectonic province. To the south, the Piedmont is covered by Cretaceous Coastal Plain sediments (the Potomac Formation). The crystalline rocks consist of the Wissahickon Formation, the Cockeysville Marble (parts of the Glenarm Series), and the Wilmington Complex. The Cockeysville Marble was not sampled. The Wissahickon Formation is in amphibolite facies, and the Wilmington Complex is in granulite facies. The metamorphism has been dated radiometrically as about 440 Ma (Grauert and Wagner, 1975) and therefore has traditionally been ascribed to the Taconic orogeny.

The Wissahickon Formation consists of gneiss, schist, and amphibolite. The major constituents of the gneiss are biotite, quartz, and feldspar. The gneiss was sampled at sites 14–19 (Fig. 2). The schist consists mainly of biotite, almandine, feldspar, and sillimanite. The sillimanite and feldspar content is much higher in the western part of the state (Ward and Groot, 1957), and so the samples collected there (site 34 in Fig. 2) are differentiated from the others (sites 1–9 and 31–33 in Fig. 2). The contact between the schist and the gneiss in the Wissahickon Formation is gradational, and the two rock types can be found interlayered with each other, although not commonly so. The amphibolite was not sampled.

The Wilmington Complex consists of anorthosite, noritic

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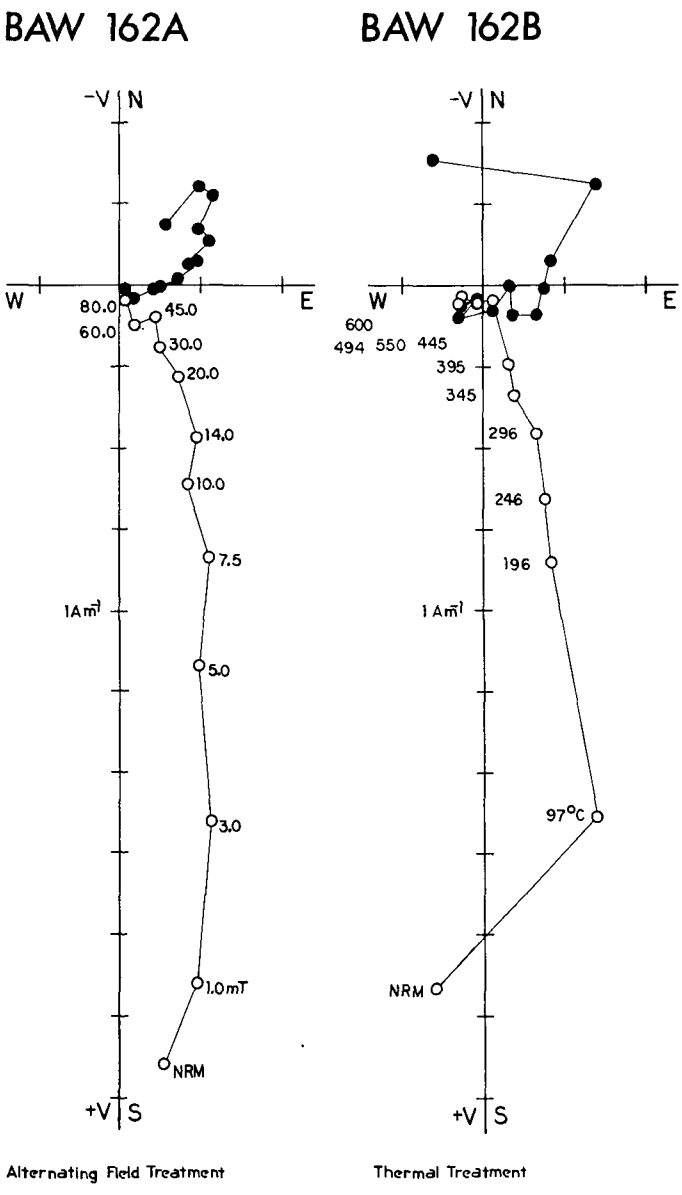


Figure 3. Orthogonal demagnetization diagrams of two specimens cut from the same sample core. Filled circles represent projections on the horizontal plane; open circles, those on the vertical plane. BAW 162A was treated with alternating-field demagnetization; BAW 162B, with thermal demagnetization. Both show a steep, downward direction.

anorthosite, norite, and charnockite, which together make up the Arden pluton and Ward's (1959) "Bringhurst Gabbro," as well as amphibolite, gabbro, and three types of gneiss. Rao and Van der Voo (1980) studied the anorthosites of the Arden pluton. In the present study, the norite in the Bringhurst area was sampled (sites 25–28 in Fig. 2). The hypersthene-quartz-andesine gneiss (Woodruff and Thompson, 1975) found in the eastern part of the complex, which is here called Wilmington Gneiss A, was not sampled. The hornblende-plagioclase gneiss with some interlayered amphibolite and schist (Woodruff and Thompson, 1975) found in the western part of the complex is here called Wilmington Gneiss B. The gneiss was sampled at sites 10–13 (Fig. 2), the amphibolite at sites 29 and 30 (Fig. 2), and the interlayered schist at site 35 (Fig. 2). The granitic gneiss of Woodruff and Thompson (1972) and the Port Deposit Granodiorite of Ward (1959) are here called "Wilmington Gneiss C." It lies mainly in northeastern Maryland but extends about 2 km across the border into western Delaware. It consists mainly of microcline, oligoclase, quartz, and biotite. It was sampled at sites 36–40 (Fig. 2). The gabbro body at Iron Hill, south of Newark in western Delaware, was sampled at sites 20–24 (Fig. 2).

The relative ages of the members of the Wissahickon Formation and the Wilmington Complex are uncertain (Woodruff and Thompson, 1972, 1975). Woodruff and Thompson (1972) stated that the gabbro at Iron Hill clearly intrudes (and is therefore younger than) the surrounding Wilmington Complex. However, it has also been suggested that this gabbro is most likely a separate upthrusted slab (Avery A. Drake, 1982, personal commun.), and therefore its relative age is uncertain.

RESULTS

The gneiss of the Wissahickon Formation yielded the most internally consistent directions of the units sampled. "Internally consistent" means that directions cluster well within site and yield site means that are similar from site to site. The Wissahickon Gneiss responded well to both alternating-field and thermal-progressive demagnetization, which removed a steep downward direction (Fig. 3). This direction is characteristic of all Wissahickon Gneiss samples (Fig. 4), and each site has a k value greater than 16 (Table 1).

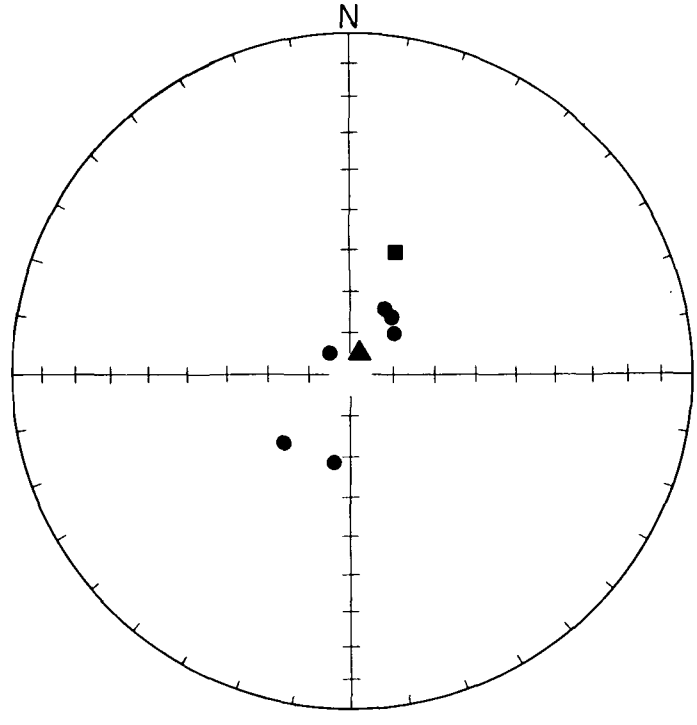


Figure 4. Equal-area stereographic projection of the site mean directions of the Wissahickon Gneiss samples (circles) and the internally consistent site from the amphibolite (square). The mean of these site means is plotted as a triangle. All directions are downward.

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There are several reasons why the directions of Wissahickon Gneiss samples are more internally consistent than those of other rocks. The intensity of magnetization is relatively strong, whereas for samples of some of the other rock units, the magnetization is so weak that it is difficult to measure even with a cryogenic magnetometer. Some other samples not only are weakly magnetized but also have a significant viscous remanent magnetization, which changes direction during laboratory storage. In addition, the Wissahickon Geniss was sampled in a deep railroad cut and a sharp road-cut that are among the freshest and best-exposed outcrops in northern Delaware. The magnetization of some samples from other units is interpreted as being affected by weathering.

Of the samples from the two sites of the amphibolite, which is interlayered with the Wilmington Gneiss B, one (site 29) exhibits good clustering (k = 32.48, $alpha_{95}$ = 10.75°), whereas the other (site 30) has no consistency at all in its measured directions (Table 1). The mean direction for the samples from site 29 is similar to site means of the Wissahickon Gneiss and is plotted together with them in Figure 4. The mean direction calculated from the site means of the amphibolite and the Wissahickon Gneiss samples is declination = 17.7°, inclination = 85.6°, k = 15.31, $alpha_{95}$ = 15.9°; it is also plotted on Figure 4.

Interpretation of the results of the samples of the schist of the Wissahickon Formation is problematic. Within each site, the clustering of sample directions is poor, yet the mean direction of each site is consistent from site to site. The site mean directions of most Wissahickon Schist samples are steeply downward, which agrees with directions of the Wissahickon Gneiss, the Wilmington Gneiss B amphibolite, and the Arden pluton. Those samples with more shallow site mean directions are generally those that cluster more poorly (that is, sites 2 and 4).

Do the Wissahickon Schist directions have a significant concentration, and, if so, what is the mean direction of that concentration? To test the significance of the concentration of Wissahickon Schist directions, all directions determined have been plotted on an equal-area stereographic projection, and the results have been con-

TABLE 1. SUMMARY OF SITE MEAN PALEOMAGNETIC DIRECTIONS

Site	*Rock unit*	*Site mean*	*k, alpha$_{95}$*	*n/N*
1	Wissahickon Schist	106.6/+75.8	11.65, 18.5°	7/14
2	Wissahickon Schist	232.0/−24.6	2.00, 54.3°	8/11
3	Wissahickon Schist	89.0/+68.8	3.42, 30.8°	10/12
4	Wissahickon Schist	236.2/ +8.5	3.42, 26.5°	13/13
5	Wissahickon Schist	43.8/+63.6	15.88, 11.8°	11/11
6	Wissahickon Schist	216.2/+79.4	5.10, 17.0°	18/19
7	Wissahickon Schist	175.2/+70.6	3.89, 24.2°	13/13
8	Wissahickon Schist	141.6/+25.5	4.33, 41.7°	5/ 7
9	Wissahickon Schist	79.8/+41.1	2.45	3/ 6
10	Wilmington Gneiss B	239.6/+36.0	131.62, 5.3°	7/ 8
11	Wilmington Gneiss B	176.4/+30.0	65.56, 6.9°	8/ 8
12	Wilmington Gneiss B	356.0/+36.1	17.64, 11.8°	10/10
13	Wilmington Gneiss B	354.9/+62.1	29.83, 7.7°	13/13
14	Wissahickon Gneiss	304.7/+83.0	59.21, 5.4°	13/14
15	Wissahickon Gneiss	34.5/+74.3	77.25, 5.2°	11/13
16	Wissahickon Gneiss	192.3/+67.1	16.26, 11.1°	12/12
17	Wissahickon Gneiss	47.1/+77.2	33.12, 6.7°	15/15
18	Wissahickon Gneiss	30.0/+73.4	51.10, 6.5°	11/11
19	Wissahickon Gneiss	224.1/+65.6	96.39, 4.2°	13/13
20	Iron Hill Gabbro, soft direction	193.1/−19.5	112.86, 4.3°	11/14
	Iron Hill Gabbro, hard direction	189.2/+58.5	23.80, 8.3°	14/14
21	Iron Hill Gabbro, soft direction	259.5/+57.7	29.23, 17.3°	4/ 7
	Iron Hill Gabbro, hard directon	197.0/+47.6	8.98, 21.3°	7/ 7
22	Iron Hill Gabbro, soft direction	90.3/ +5.7	257.24, 7.7°	3/ 3
	Iron Hill Gabbro, hard direction	61.6/+42.5	2.79	3/ 3
23	Iron Hill Gabbro, soft direction	353.6/+82.9	3.85, 28.4°	10/12
	Iron Hill Gabbro, hard direction	187.5/−45.6	21.51, 14.8°	6/12
24	Iron Hill Gabbro, soft direction	18.7/ −2.5	14.86, 20.5°	5/ 6
	Iron Hill Gabbro, hard direction	32.9/−15.9	62.84, 8.5°	6/ 6
25	Norite	159.1/+86.3	146.64, 3.6°	12/12
26	Norite	185.1/ +6.2	37.71, 11.1°	6/ 9
27	Norite	55.4/−19.5	55.28, 16.7°	3/ 3
28	Norite	226.5/−49.7	8.54, 45.1°	3/ 3
29	Gneiss B, interlayered amphibolite	19.5/+60.2	32.48, 10.8°	7/12
30	Gneiss B, interlayered amphibolite	283.3/−12.8	6.31, 33.0°	5/ 5
31	Wissahickon Schist			0/ 1
32	Wissahickon Schist	7.8/+52.2	53.68, 12.7°	4/ 4
33	Wissahickon Schist			0/ 2
34	Wissahickon Schist, western facies	288.7/+22.7	81.27, 7.5°	6/ 6
35	Gneiss B, interlayered schist	264.7/+40.5	9.23, 43.1°	3/ 3
36	Wilmington Gneiss C	358.9/+35.3	5.23, 20.1°	13/14
37	Wilmington Gneiss C	174.4/+41.5	218.10, 6.2°	4/ 4
38	Wilmington Gneiss C	3.5/+84.4	5.96, 17.8°	14/17
39	Wilmington Gneiss C			0/ 1
40	Wilmington Gneiss C	216.2/+42.7	144.06, 23.6°	2/ 3

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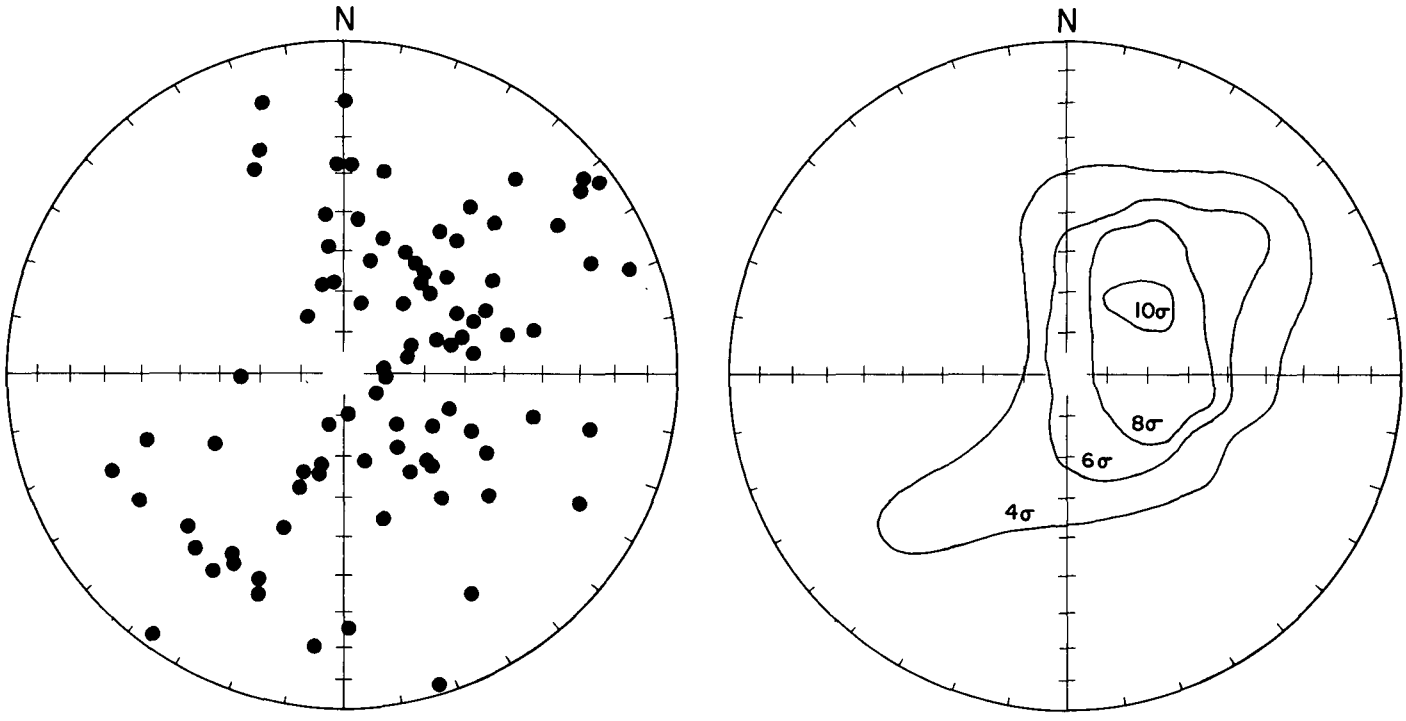


Figure 5. Equal-area projection of all specimen directions determined for the Wissahickon Schist (left). All directions are downward (upward directions have been inverted). Contour diagram of the data (right), using the Kamb (1959) method.

toured, using the Kamb (1959) method. A direction is plotted for each specimen of the Wissahickon Schist that was demagnetized, except those for which the demagnetization diagram was uninterpretable. The contour diagram (Fig. 5) shows that the Wissahickon Schist directions are not random. They show a significant concentration around a steep, downward, northeasterly direction (declination = 46°, inclination = + 66°). The concentration is greater than 10 *sigma* using Kamb's statistics, or more than three times the concentration that would result from a uniform distribution. The contour diagram also shows that the scattering that is present is not systematic. The distribution of directions is unimodal. The significant steep direction is outlined by an equant contour, indicating that it is not formed by a smear of directions between end members, and there is no other significant concentration of directions. Therefore, the steep direction is unlikely to be the vector sum of two or more ancient directions concurrently removed during progressive demagnetization.

It is not clear why the Wissahickon Schist directions are so poorly clustered within each site. One possibility is that there was a small amount of random rotation after the acquisition of magnetization on a scale such that samples collected within one metre or less of each other were rotated different amounts in different directions. This would account for the scatter observed in the magnetic directions. No specific mechanism is proposed, but it is easy to imagine such rotations taking place in schist that is deformed on a very small scale. Another possibility is that weathering scattered the magnetic directions. Samples from sites 3–9 and 31–33 are visibly weathered. It is always preferable to sample fresh outcrop, but in the Piedmont Province of northern Delaware, outcrop is scarce, and one must sample whatever is available. Ellwood (1981) has shown that weathering can randomly scatter a consistent remanent magnetization by the formation of hematite and by maghemitization of the original magnetite. If, in contrast to Ellwood's samples, the coercivities or blocking temperatures of the original magnetization in our samples are similar to those of the magnetization due to weathering, then the two magnetizations are removed simultaneously in progressive demagnetization, and the resulting directions would be scattered in a random manner.

The results from the other units sampled (the gabbro, norite, and interlayered schist of the Wilmington Complex, the Wilmington Gneisses B and C, and the western facies of the Wissahickon Schist) are uninterpretable. In general, samples respond well to progressive demagnetization but yield widely divergent directions (Fig. 6). The clustering within site is often very good (Table 1), but samples from all sites, even samples from sites within the same rock unit, yield different magnetic directions. Figure 7 shows the mean directions of all sites with good internal clustering of directions (k greater than 15), along with directions corresponding to representative Phanerozoic paleopoles for cratonic North American (Van der Voo, 1981). It can be seen that there is no clustering of these site means. The greatest concentration is in the southwest quadrant; whatever the cause of these scattered directions, it is unlikely that they represent North American remagnetizations.

It is clear that for some reason, the last-described rock units did not serve as accurate recorders of the ancient magnetic field. Why, then, do they behave so well when subjected to progressive demagnetization and why do the directions commonly cluster within site? For the gabbro at Iron Hill and the norite, the unverifiable answer may be that some large boulders of float were sampled rather than outcrop. During sample collection, sites 20–22, 25, and 27 (Fig. 2) were noted as being located at questionable outcrop. Another possibility for the gabbro, if it is intrusive into the sur-

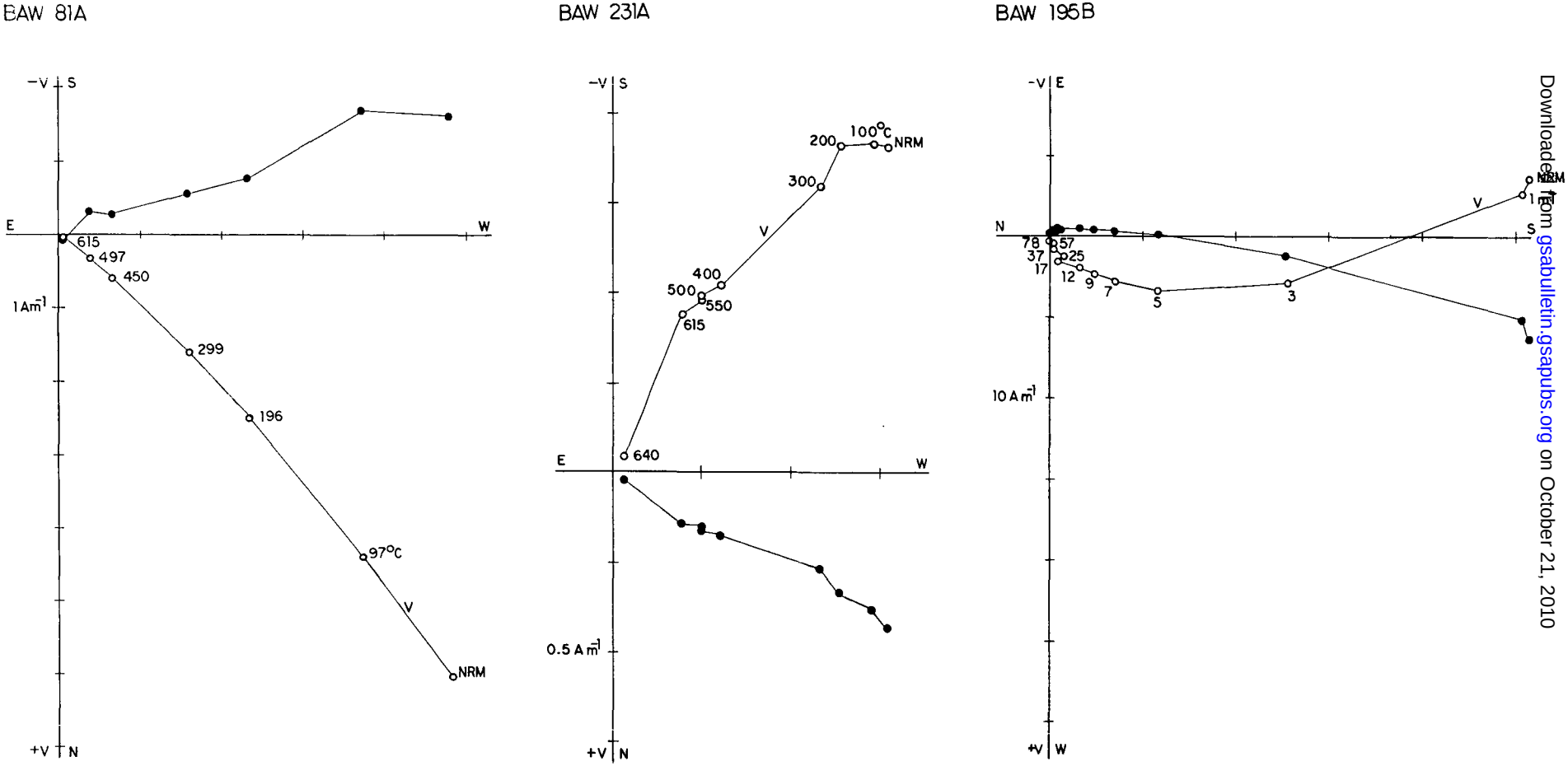


Figure 6. Orthogonal demagnetization diagrams (as in Fig. 3), plotted with axes in different orientations for clarity. BAW 81A, from Wilmington Gneiss B (thermal treatment), shows a moderately downward, west-southwesterly direction. BAW 231A, from the norite (thermal treatment), shows a moderately upward, west-northwesterly direction. BAW 195B, from the Iron Hill Gabbro (alternating-field treatment), shows two components: a shallow, westerly component is removed up to 17 mT, leaving a steeply downward component.

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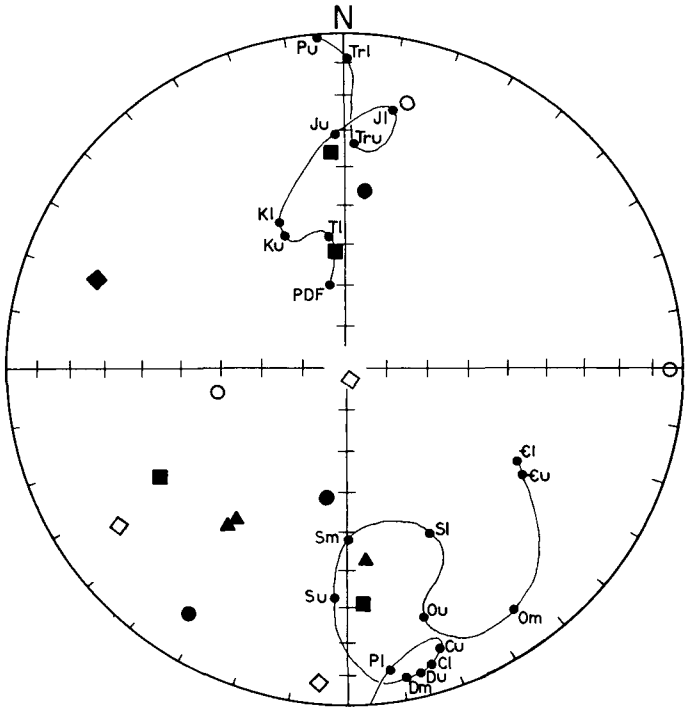


Figure 7. Equal-area stereographic projection of the mean directions of all sites with good internal clustering from the norite (open diamonds), gabbro (high-coercivity–high-blocking temperature directions are filled circles, low-coercivity–low-blocking temperature and directions are open circles), Wilmington Gneisses B (squares) and C (triangles), and the western facies of the Wissahickon Schist (filled diamond), along with Phanerozoic paleomagnetic data from cratonic North America plotted as expected field directions relative to northern Delaware. All directions are downward (upward directions have been inverted for clarity).

rounding Wilmington Complex as Woodruff and Thompson (1972) suggest, is that emplacement postdates the Taconic metamorphism, and thus the magnetization is younger than Ordovician. If the gabbro is a separate obducted slab or fragment, then the magnetization would also be different from that of the Wilmington Complex and Wissahickon Formation (Avery A. Drake, 1982, personal commun.). In any event, the magnetization is not consistent, as Figure 7 shows.

For the other samples from the sites in the Wilmington Complex gneisses and in the western facies of the schist of the Wissahickon Formation that have good internal clustering of directions (site means are plotted on Fig. 7), these explanations are not viable. These units were definitely involved in the 440-Ma metamorphism, and they were sampled at reliable outcrops. Figure 7 shows that none of these directions corresponds to any Phanerozoic North American direction, and so remagnetization of the rocks as part of North America is not a possible explanation. Interpretation of these directions as post-Ordovician African directions is a possibility, but any such interpretation would be highly speculative.

CONCLUSIONS

The Wissahickon Gneiss and the amphibolite interlayered in the Wilmington Gneiss B yield a paleopole located at 48°N, 72°W. The Wissahickon Schist has a significant concentration of magnetizations around a direction that corresponds to a pole position of 56°N, 16°W. These pole positions are shown in Figure 8, along with Ordovician pole positions for cratonic North America and the mean British Ordovician pole (Briden and others, 1973; Faller and others, 1977), relocated by correcting for the opening of the Atlantic (Bullard and others, 1965). The pole positions from Delaware lie very close to the study area and are significantly divergent from other North American poles of the same age. Thus, when this part of the Piedmont was metamorphosed at about 440 Ma, it was most likely associated with some other plate across the Iapetus Ocean. The preferred interpretation is that the study area was associated with Gondwana rather than Europe, because the Delaware poles are divergent from the British pole. Few reliable Ordovician poles have been determined for Africa, and so this hypothesis cannot be tested in great detail. However, the few available paleomagnetic data and the geologic evidence for Ordovician-Silurian glaciations in western Gondwana (summarized by Crowell, 1981) indicate that northern Africa was located at very high latitudes during the Late Ordovician. Thus, if the Piedmont of northern Delaware was associated with northwestern Africa when it was metamorphosed 440 Ma ago, the magnetization acquired would be very steeply inclined, which is indeed the case.

No conclusions can be drawn from the data from the other rock units sampled in northern Delaware because the directions found are inconsistent at the rock-unit level. However, the directions found do not correspond to Phanerozoic North American poles, and so the conclusions drawn from the Wissahickon Schist,

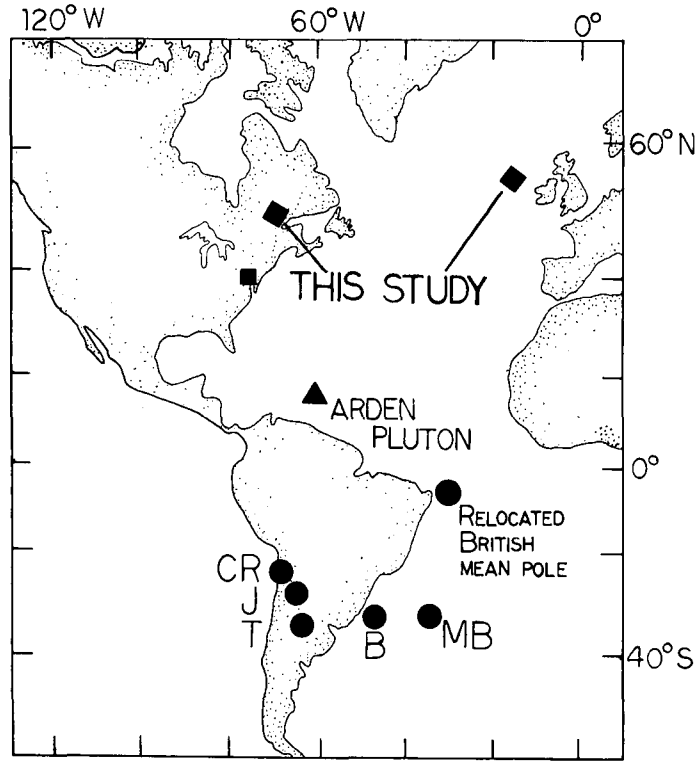


Figure 8. Map showing the site location (squares) and Late Ordovician pole positions for this study (diamonds). Poles for cratonic North America (filled circles) are: Chapman Ridge Formation (CR) and Moccasin-Bays Formations (MB) (Watts and Van der Voo, 1979); Juniata Formation (J) (Van der Voo and French, 1977); Trenton limestone (T) (McElhinny and Opdyke, 1973); Beemerville Complex (B) (Proko and Hargraves, 1973).

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the Arden pluton, the Wissahickon Gneiss, and the amphibolite interlayered in the Wilmington Gneiss B are not refuted.

The consistent magnetic directions found in this study and the directions determined for the Arden pluton are all very steeply inclined. The Arden pluton directions are upward, whereas most directions from the present study are downward. The agreement of the directions from sampling sites located all across northern Delaware indicates that it is unlikely that the steep direction is the result of local tilting of the rocks sampled. It is very unlikely that such a large area would tilt as a unit. Our interpretation of this steep direction agrees with that of Rao and Van der Voo (1980). The preferred explanation is that the Piedmont of northern Delaware formed as part of Gondwana, became sutured to North America when the Iapetus Ocean closed, and remained attached to North America when the Atlantic Ocean opened. Although the metamorphism is 440 Ma, if our interpretation of the paleomagnetic data is correct, it cannot be called Taconic, because it occurred while the terrane was associated with Gondwana, and the Taconic is a North American event. The collision that sutured the Piedmont to North America must have occurred during a later orogeny, either the Acadian or the Alleghenian.

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REFERENCES CITED

Briden, J. C., Morris, W. A., and Piper, J.D.A., 1973, Paleomagnetic studies in the British Caledonides—VI, Regional and global implications: Royal Astronomical Society Geophysical Journal, v. 34, p. 111–121.

Bullard, E. C., Everett, J., and Smith, A. G., 1965, The fit of the continents around the Atlantic, *in* A symposium on continental drift: Royal Society of London Philosophical Transactions, ser. A, v. 258, p. 41–51.

Crowell, J. C., 1981, Early Paleozoic glaciation and Gondwana drift, *in* McElhinny, M. W., and Valencio, D. A., eds., Paleoreconstruction of the continents: Washington, D.C., American Geophysical Union; and Boulder, Colorado, Geological Society of America, p. 45–49.

Ellwood, B. B., 1981, Weathering effects on the magnetic properties of the Milledgeville granite, Georgia: Earth and Planetary Science Letters, v. 55, p. 311–316.

Faller, A. M., Briden, J. C., and Morris, W. A., 1977, Paleomagnetic results from the Borrowdale volcanic group, English Lake District: Royal Astronomical Society Geophysical Journal, v. 48, p. 111–121.

Grauert, B., and Wagner, M. E., 1975, Age of the granulite facies metamorphism of the Wilmington Complex, Delaware-Pennsylvania Piedmont: American Journal of Science, v. 275, p. 683–691.

Kamb, W. B., 1959, Ice petrofabric observations from Blue Glacier, Washington, in relation to theory and experiment: Journal of Geophysical Research, v. 64, p. 1891–1909.

McElhinny, M. W., and Opdyke, N. D., 1973, Remagnetization hypothesis discounted: Paleomagnetic study of the Trenton Limestone, New York State: Geological Society of America Bulletin, v. 84, p. 3697–3708.

Proko, M. S., and Hargraves, R. B., 1973, Paleomagnetism of the Beemerville (New Jersey) Alkaline Complex: Geology, v. 1, p. 185–186.

Rankin, D. W., 1975, The continental margin of eastern North America in the south Appalachians: The opening and closing of the proto-Atlantic Ocean: American Journal of Science, v. 275A, p. 298–336.

Rao, K. V., and Van der Voo, R., 1980, Paleomagnetism of a Paleozoic anorthosite from the Appalachian Piedmont, northern Delaware: Possible tectonic implications: Earth and Planetary Science Letters, v. 47, p. 113–120.

Van der Voo, R., 1981, Paleomagnetism of North America—A brief review, *in* McElhinny, M. W., and Valencio, D. A., eds., Paleoreconstruction of the continents: Washington, D.C., American Geophysical Union; and Boulder, Colorado, Geological Society of America, p. 159–176.

Van der Voo, R., and French, R. B., 1977, Paleomagnetism of the Late Ordovician Juniata Formation and the remagnetization hypothesis: Journal of Geophysical Research, v. 82, p. 5796–5820.

Wagner, M. E., and Crawford, M. L., 1975, Polymetamorphism of the Precambrian Baltimore Gneiss in southeastern Pennsylvania: American Journal of Science, v. 275, p. 653–682.

Ward, R. F., 1959, Petrology and metamorphism of the Wilmington Complex, Delaware, Pennsylvania, and Maryland: Geological Society of America Bulletin, v. 70, p. 1425–1458.

Ward, R. F., and Groot, J. J., 1957, Engineering materials of northern New Castle County: Delaware Geological Survey Bulletin No. 7, 103 p.

Watts, D. R., and Van der Voo, R., 1979, Paleomagnetic results from the Ordovician Moccasin, Bays, and Chapman Ridge Formations of the Valley and Ridge Province, eastern Tennessee: Journal of Geophysical Research, v. 84, p. 645–655.

Woodruff, K. D., and Thompson, A. M., 1972, Geology of the Newark area, Delaware: Delaware Geological Survey Geologic Map Series, No. 3.

—— 1975, Geology of the Wilmington area, Delaware: Delaware Geological Survey Geologic Map Series, No.4 .

Zijderveld, J.D.A., 1967, AC demagnetization of rocks: Analysis of results, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 254–286.

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